SD-28

Standardization Decisions for a Modular Open Systems Approach (MOSA)



DEFENSE STANDARDIZATION PROGRAM OFFICE

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Foreword

The world we live in today requires faster delivery of capabilities to the warfighter to keep pace with emerging threats. As contested environments become more complex due to multi-domain operations and rapid technology innovation by foreign adversaries, the development and sustainment of weapon systems necessitate modernized thinking and approaches to ensure the United States remains dominant. The Department of War (DoW) recognizes the use of a modular open systems approach (MOSA) as the preferred method for designing affordable and adaptable systems. The Department's Acquisition Transformation Strategy exemplifies this by stating that implementing MOSA is "paramount for interoperability, competition, technology insertion, performance improvements, and reducing life cycle costs."

As the DoW furthers the use of MOSA in all warfighting capabilities to the maximum extent practicable, we have come to realize that standards are an essential element to its implementation. To this end, the Acquisition Transformation Strategy further directs that the Department "will fully implement MOSA in all acquisition strategies to the maximum extent practicable by leveraging open standards, pursuing software application programming interfaces, and asserting IP rights."

DoW acquisition and technical professionals (e.g., program managers, MOSA practitioners, engineers, and support contractors) must develop the knowledge and understanding of how standardization and standards development can be used to achieve the benefits of MOSA. The Standardization-related Document (SD)-28 identifies best practices and concepts that help these professionals make smarter and more deliberate decisions when implementing a MOSA. This publication is a result of extensive research and collaboration through the Modular Open Systems Working Group (MOSWG) Standards Tiger Team.

Recommended changes to this document should be addressed to the Defense Standardization Program Office, 8725 John J. Kingman Road, Stop 5100, Fort Belvoir, VA 22060-6220, or via email to DSPO@dla.mil.



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Section 1. Introduction

1.1 Background

The use of a modular open systems approach (MOSA) in DoW weapon systems is imperative to ensure the integration of technological innovation to meet the needs of the warfighter at the speed of necessity. Standards, as key enablers in the implementation of a MOSA, provide the foundation to ensure that technology is safe, universally adopted, and interoperable, and are interwoven into the five MOSA principles:

1. Establish an enabling environment.
2. Employ a modular design.
3. Designate key interfaces.
4. Use widely used consensus-based standards.
5. Certify conformance.

In a MOSA, standards are used to develop capabilities and to define hardware and software interfaces between systems, components, and/or modules. This approach is advantageous for the DoW and defense contractors. Defense contractors can retain proprietary information regarding the inner workings and/or design—the “how” of a subsystem—while the government has full knowledge of the included functionality—the “what” of a subsystem (i.e. the black box theory). With clearly defined interfaces supporting the encapsulated functionality, the DoW can accept bids on future contracts without fear of being locked in with a specific vendor for the sustainment of a system or subsystem. A MOSA also enables market competition and scalability of production through multiple producers, and the incremental insertion of newer and refreshed capabilities to improve reliability and maintainability, reduce costs, and add capability. Likewise, conformance to interface standards ensures implementations are consistent with the standard and can assist with information sharing across domains.

Figure 1 shows a notional example of the application of standards to help achieve the benefits of MOSA with a hypothetical modularized DoW weapon system built by Vendor A. In the system architecture, the navigation system was determined to be a module, and its logical interface is well defined by a standard, ISO/IEC 17267, *Intelligent transport systems — Navigation systems — Application programming interface (API)*. As a result of this approach, the government has the flexibility to procure and integrate a new navigation system from another offeror (e.g., Vendor B) with equivalent or advanced capabilities without major system overhaul. This can reduce maintenance costs and issues associated with vendor lock or proprietary technology. Realistically, in today’s complex systems, software must often comply with multiple standards (e.g., functional, interoperability, syntactic, semantic, operational, and protocols), and while the selection of a standard like ISO/IEC 17267 may be the correct choice for a particular MOSA, it may not suffice for another MOSA.

Adequate resourcing and investment in the development, selection, and implementation of standards are essential for the DoW to drive the future of technological development and implement a MOSA in warfighting systems.

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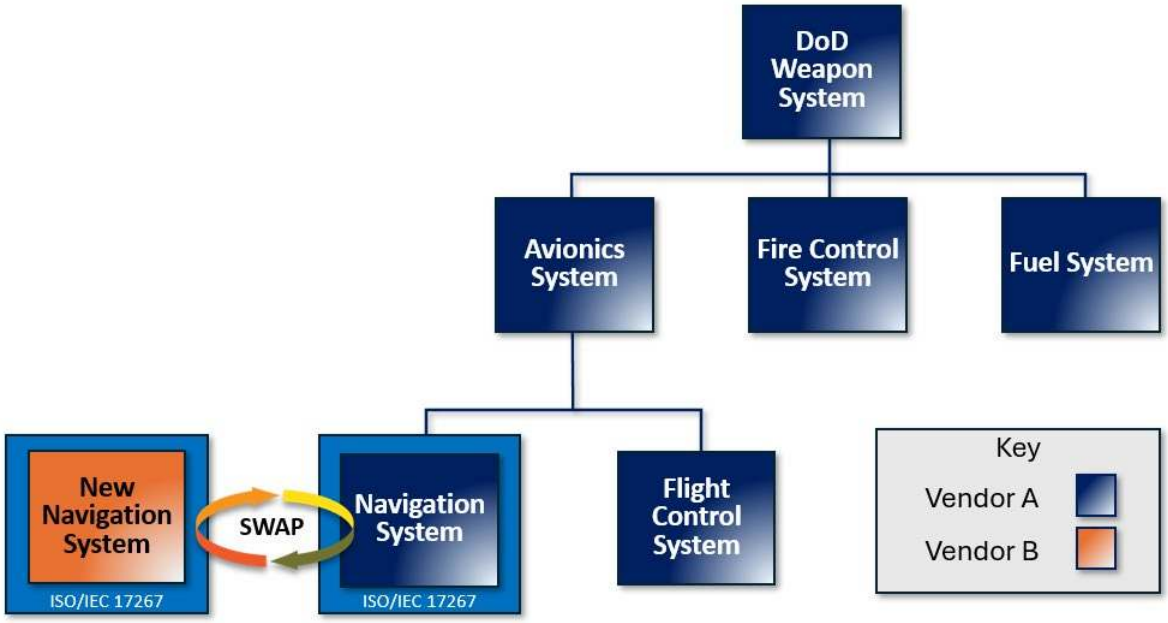


Figure 1. Example Breakdown of DoW Weapon System Architecture

1.2 Purpose

This publication provides guidance to government and industry personnel on making standardization decisions to support a MOSA. This guidance is intentionally non-prescriptive.

1.3 Definitions

Statutory requirements for implementing MOSA can be found in 10 United States Code (U.S.C.) §4401-4403:

- 10 U.S.C. §4401, Requirements for MOSA in Defense Acquisition Programs and Definitions
- 10 U.S.C. §4402, Address MOSA in Program Capabilities Development and Acquisition Weapon System Design
- 10 U.S.C. §4403, Availability of Major System Interfaces and Support for MOSA.

Within the statute, there are several terms identified as relevant to implement a MOSA. While many of these terms are already defined, some important terms for discussing standardization are not. Likewise, some terms that are commonly used within MOSA stakeholder communities have developed unique meanings that require definitions to provide clarity and to prevent confusion. All terms related to standardization that are not already defined in 10 U.S.C. §4401 will be defined in this section to support further discussion.

10 U.S.C. §4401 directs that DoW acquisition programs use widely supported and consensus-based standards to enforce modular system interfaces (MSIs), if such standards are available and suitable to use. This publication interprets the phrase “widely supported and consensus-based standards” to be inclusive of either or both conditions to promote practical implementation of a MOSA. The following sections define both terms in that phrase so stakeholders can differentiate between the two.

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1.3.1 Widely supported

In the context of standardization, the term **widely supported** can be attributed to following three characteristics:

- **Adoption**. There are three methods for adoption of widely supported standards:
  - The standard has been adopted by the DoW. This can include government (e.g., defense standards) and non-government standards (e.g., standards developed by standards development organizations (SDOs) or civil standards bodies). ASSIST is the official source of defense and federal standards and specifications and includes adoption notices for non-government standards adopted for use by the DoW.
  - The standard has been formally recognized by an authoritative source (e.g., the Modular Open Systems Working Group (MOSWG) or SDOs).
  - The standard is a de facto standard.
- **Implementation/Use**. A preponderance of current and emerging programs or systems is implementing products or systems in adherence to the standard.
- **Stakeholders**. A preponderance of stakeholders in the respective technical domain advocates for use of the standard.

A standard that satisfies at least one of the above-mentioned characteristics can be deemed widely supported.

1.3.2 Consensus-Based

As defined in Office of Management and Budget (OMB) Circular A-119, the term **consensus-based** is defined as “general agreement, but not necessarily unanimity. During the development of consensus, comments and objections are considered using fair, impartial, open, and transparent processes.”

Consensus can be obtained in a variety of ways, including various voting structures (e.g., one vote per individual, organization/company, or country/government). In the context of standards and standards development, gathering consensus is needed to adjudicate feedback to standard revisions or filter ideas and scope for developing a standard.

1.3.3 Open Standard or Specification

The term **open standard or specification** is defined as a standard or specification accessible to those in the domain of interest. This can include widely supported or consensus-based standards, which are preferred for use. The accessibility of a standard refers to both access to the standard and the openness of the standards development organization to allow for meaningful engagement at all stages of standards development. A standard’s accessibility is determined by the licensing structure for the standard, and openness is understood as fair, reasonable, and non-discriminatory terms and fees.

Open standards and specifications are not synonymous with free use. In the context of standardization, the term *open* refers to the accessibility of underlying technology. The details of the configuration that can be attributed to the work of the company, team, or individual are subject to their financial strategies, such as free-use or paid license.

1.3.4 MOSA-Enabling Standard or Specification

The term **MOSA-enabling standard or specification** refers to a standard or specification that supports the technical or business implementation of strategic and program MOSA objectives. Implementation of MOSA-enabling standards or specifications alone does not make a program achieve a MOSA. However, by implementing one or more MOSA-enabling standards, along with a modular architecture and

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appropriate technical data and licensing rights, a system may be able to achieve one or more of the primary objectives of MOSA:

- Cost savings or avoidance (from modularity decisions)
- Schedule reduction (from modularity decisions)
- Technology refresh and upgradability
- Interoperability, including system of systems interoperability and mission integration
- Other benefits during the sustainment phase, such as reduced training requirements, fewer depot-level supplies, simpler troubleshooting, and more rapid “flight line” repairs.

MOSA-enabling standards or specifications can be categorized as widely supported, consensus-based, open, or a combination. They can include software interface standards or application programming interfaces (APIs) that provide a set of rules and specifications that enable different software applications to interact by facilitating data exchange (i.e., a method for communication and data exchange). As noted in 10 U.S.C. §4401, information regarding interfaces of interest for DoW warfighting systems can be found in the Military Department interface repositories.



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Section 2. Openness for Standards and Specifications

2.1 Openness as a Spectrum

In the context of standards and specifications, the term *open* is mistakenly used synonymously with free and publicly available. Openness is not a binary, to mean that standards and specifications are either open or closed. Openness is a spectrum of possibilities, driven by licensing and stakeholder participation structures, as shown in Figure 2.

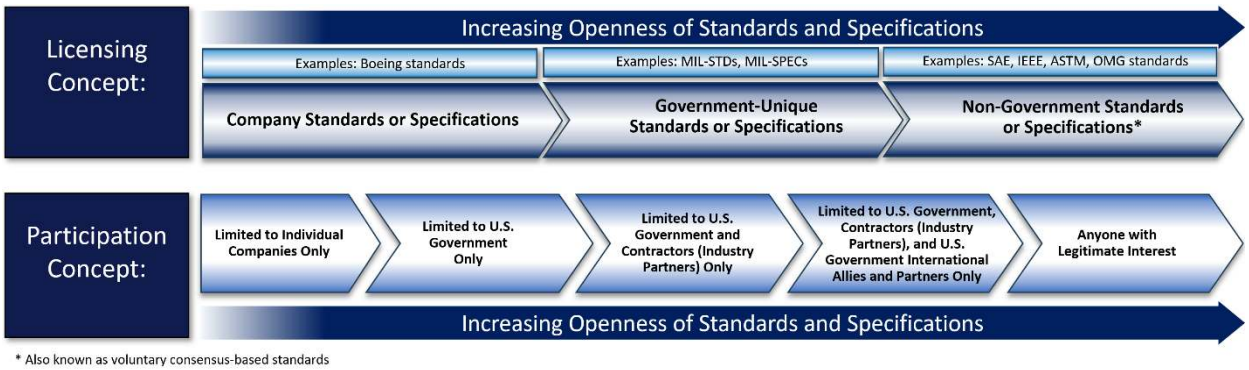


Figure 2. The Spectrum of Openness for Standards and Specifications

Figure 2 illustrates the concept of openness and the range of possibilities. It shows how different types of standards and specifications can be used and the pros and cons associated with each use. The desired level of openness is determined by the stakeholder community that can develop (i.e., participating concept) and access (i.e., licensing concept) the standard or specification.

Figure 2 applies in most cases but is not comprehensive. Despite considering many different licenses and participation schema, no single model can cover all possibilities because SDOs freely manage their membership and intellectual property (IP). Similarly, IP licensing and delivery mechanisms for standardization content may continue to change as SDOs adopt new forms of digital content.

2.1.1 Licensing Concept

Licensing refers to the ability for users to acquire and use authorized copies of standards and specifications of interest. Licensing structures typically range from those dictated by a specific organization to those developed by SDOs. Openness increases as the licensing structure expands and the standards and specifications become more accessible to users, as shown in Figure 2.

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The following list explains each type of licensing structure in detail, proceeding in order of increasing openness:



- **Company Standards and Specifications**. Companies that develop standards and specifications define the associated licensing restrictions. Access to these standards and specifications can be difficult because the company can restrict access to anyone at its discretion, including stakeholders outside the company or even specific directorates or teams within the company. Examples of company standards and specifications can include those developed by defense contractors, such as Boeing or Lockheed Martin.
- **Government-Unique Standards and Specifications**. Standards and specifications developed by the U.S. Government (USG) do not contain typical organization licensing restrictions. Rather, the USG has dissemination controls for defense and space-related articles and technical data, which may limit access to government-unique standards and specifications. Distribution statements and classification markings are utilized to further distinguish access applicability to certain stakeholders, preventing spillage and unintentional disclosure of information that can impact national security. Examples of government-unique standards and specifications include defense standards (MIL-STDs) and specifications (MIL-SPECs), which are developed solely to meet military-unique requirements and are owned by the DoW.
- **Non-Government Standards**. Non-government standards (NGSs) are documents developed by the private sector through a consensus-based standards body, government-industry consortia, or SDO. Examples of NGS include the standards developed by SAE International, ASTM International, and Object Management Group. The U.S. Government adopts NGSs, when available and practical, in lieu of developing their own standards. NGSs are usually copyrighted and made available under a license, often for a fee (though not always). The license grants access to the NGS and describes limitations such as who is granted access to the IP, how long the access is granted, and the authorized uses of the IP.

2.1.2 Participation Concept

Participation refers to stakeholder participation in developing and managing standards and specifications. Participation typically ranges from those within an individual company to anyone with legitimate interest, as shown in Figure 2. Openness increases as participation becomes more inclusive, allowing more users to develop and manage standards and specifications. The following list explains each type of participation in detail, proceeding in order of increasing openness:



- **Limited to Individual Companies Only**. Participation in standards development and management is limited to those within an individual company. Within a company, further restrictions can be placed to limit participation within specific teams or directorates to meet organizational technical or business objectives associated with standards development and management, including the protection of IP.
- **Limited to U.S. Government Only**. Stakeholder participation is limited to those within the USG, including government civilian and military personnel. This degree of participation (openness) does not include USG contractors. OMB Circular A-119 provides additional guidance for agencies participating in SDOs.
- **Limited to U.S. Government and Contractors**. Government civilian, military, and contractor personnel can participate in the development and management of standards and specifications.
- **Limited to U.S. Government, Contractors, and International Allies and Partners**. Standards collaboration with international allies and partners is imperative to increase operational effectiveness and enhance interoperability in support of joint operations. The North Atlantic Treaty Organization (NATO) and ABCANZ Armies are examples of multi-national treaty organizations in which member nations (including the U.S. and allies and partners) can come together to achieve national and international strategic objectives.
- **Anyone with legitimate interest**. Stakeholder participation can include anyone with legitimate interest in the standards and specifications. This can include personnel from industry and academia, or consumer groups and the public. This category has much relevance and interest related to open source and software interface and API applications.

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2.1.3 *Protecting Intellectual Property*

Discussions about openness must also consider how to implement appropriate security measures to safeguard IP within standards and specifications. These discussions should address cybersecurity and user and stakeholder access controls.



- **Cybersecurity**. Cybersecurity refers to the protection of computer and communication systems to ensure information within is trusted, available, and has integrity. Cybersecurity measures should be implemented in accordance with DoDI 8500.01 to help prevent attack of systems infrastructure to garner information illegally.



- **User Access Controls**. User access controls refer to methods that ensure only authorized users obtain access to standards and specifications. Implementing access controls requires systems for accessing standards and specifications to have appropriate security enclaves so only those with the need for legitimate access can access the information contained within the system. To determine whether users have a legitimate need for access, the agency granting access must obtain information from the users to determine their need to know and must assign appropriate personnel to evaluate the information provided. The following non-exhaustive list provides examples of the information that may be needed to establish user access controls.
  - **Citizenship Status**. To determine if the individual is a foreign national
  - **Government Sponsor POC**. To confirm if the individual is working an active government contract
  - **Email address**. To determine if the individual is affiliated with government, private industry, or academia
  - **Purpose**. To determine if the individual has the need to know for the requested IP.



- **Stakeholder Access Controls**. Stakeholder access controls refer to the methods in which developers and management personnel have access to develop and manage standards and specifications of interest. Like user access controls, stakeholder access controls require appropriate system security and personnel verification to verify access is legitimate. For standards and specifications in development or undergoing subsequent revisions, it is especially important to ensure that information is kept confidential until the appropriate licensing structure is put in place to safeguard IP.

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Section 3. What Does "Open" Mean in the Context of MOSA?

The MOSA acronym incorporates four elements that should be satisfied in the implementation of MOSA: modular, open, systems, and approach. Each element has meaning, and 10 U.S.C. §4401 details requirements to achieve each of the elements, as illustrated in Table 1.

Table 1. 10 U.S.C. §4401 Requirements for Implementation of MOSA

Abbreviation	Requirements
Modular:	Uses a system architecture that allows severable major system components and modular systems at the appropriate level to be incrementally added, removed, or replaced throughout the life cycle of a major system platform to afford opportunities for enhanced competition and innovation while yielding five MOSA benefits
Open:	Is subjected to verification to ensure that relevant modular system interfaces i) comply with, if available and suitable, widely supported and consensus-based standards; or ii) are delivered pursuant to the requirements established in Section 804 of the Fiscal Year (FY) 2021 National Defense Authorization Act
Systems:	Employs a modular design that uses modular system interfaces between major systems, major system components and modular systems
Approach:	An integrated business and technical strategy that complies with technical data rights set forth in 10 U.S.C. §3771-3775

To achieve openness for MOSA, the statute states either of two approaches that can be used for modular system interfaces. This two-pronged approach provides the government the appropriate information needed to accept bids for modules in the future and avoid vendor lock.

3.1 Approach #1

The first approach focuses on the use of widely supported and consensus-based standards to define modular system interfaces (Figure 3). These standards are preferred because they are accessible to most standards users.

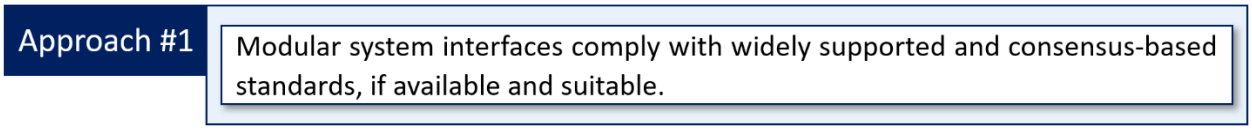


Figure 3. Approach Using Widely Supported and Consensus-Based Standards

3.2 Approach #2

The second approach allows contractors to use interface standards that are not as widely accessible or contain proprietary information if minimum data deliverables are provided to the government (Figure 4).

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Approach #2

Modular system interface data deliverables are delivered as agreed upon by the USG and Contractor, including the format and method of delivery. This applies to all relevant DoW contracts entered into after the date on which regulations and guidance are implemented (1 year after the enactment of the most recent NDAA). Required data deliverables (for modular system interfaces for modular systems deemed relevant in acquisition strategy or documentation) include:

1. Software-defined interface syntax and properties, specifically governing how values are validly passed and received between major subsystems and components.
   - Minimum Required Delivery: Machine-readable format (e.g., PDF, XML, SysML, UML)
2. A definition of the relationship between the delivered interface and existing common standards or interfaces available in Department interface repositories.
   - Minimum Required Delivery: Machine-readable format (e.g., PDF, XML, SysML, UML)
3. Functional descriptions of software-defined interfaces, conveying semantic meaning of interface elements, such as the function of a given interface field.
   - Minimum Required Delivery: Flexible

**Consider the applicability of each specific data deliverable based on the modular system interface.

Figure 4. Approach Using Data Deliverables to Ameliorate Restricted or Proprietary Information

All MSI data deliverables for modular systems deemed relevant in the acquisition strategy or documentation shall be delivered as agreed upon by the government and contractors, including the format and method of delivery. The dense statutory language has been translated into the guidance in Figure 5, providing context for the various deliverables, including format and delivery, that are required by the USG.

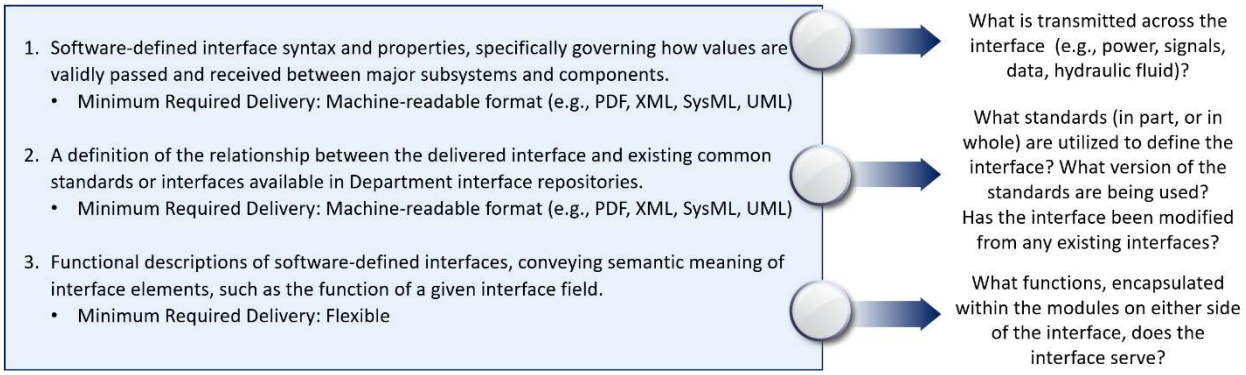


Figure 5. Deciphering 10 U.S.C. §4401 Regarding MSI Data Deliverables

If the modular system interface is not defined by widely supported and consensus-based standards, the government needs answers to the following questions about each modular system interface to be able to accept bids for future modules of interest:

1. What is transmitted across the interface (e.g., power, signals, data, hydraulic fluid)?
2. What standards (in part, or in whole) are utilized to define the interface? What version of the standards are being used? Has the interface been modified from existing interfaces?
3. What functions, encapsulated within the modules on either side of the interface, does the interface serve?

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3.2.1 Modular System Interface Data Deliverables

MSI deliverables should be provided in a way that is usable by all parties, avoiding proprietary tools or formats. 10 U.S.C. §4401 notes that two of three deliverables must be provided in a machine-readable format (e.g., Portable Document Format (PDF), Extensible Markup Language (XML), Systems Modeling Language (SysML), and Unified Modeling Language (UML)). This requirement ensures documentation is semantically unambiguous and leaves less room for poor documentation by the originator and misinterpretation by the user. Flexibility is required to ensure the appropriate format for the end deliverable. For example, user manuals would better be suited for PDF formats, while data descriptions may be better suited for XML. Including human-readable format deliverables in addition to the minimum required machine-readable format may provide additional benefits if the documentation is complete, ensures quality, and identifies missing information.

Ultimately, technical data rights for modular system interfaces must comply with 10 U.S.C. §3771-3775. DoW personnel can also refer to the guidance in the *Intellectual Property Guidebook for DoD Acquisition,* from the Office of the Under Secretary of War for Acquisition and Sustainment IP Cadre, to garner more information about MSI technical data rights.

3.3 Interfaces Without Resource Exchanges

Not all interfaces involve the exchange of resources (e.g., information, energy, fuels). For example, pure hardware interfaces do not transmit anything and are typically structural in nature, such as mount points, fasteners, couplers, bolts circles etc. In cases like this, it is important to consider the applicability of each specific data deliverable listed in Approach #2. Ideally, these types of interfaces make use of standards (e.g., Picatinny rail (MIL-STD-1913), fasteners use, standards threads). The interface documentation should include detailed descriptions of the behavioral and kinematic aspects of the interface, to understand how various components work together and communicate. Integrator instructions may be useful in understanding the applicability of these MSI deliverables.



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Section 4. Choosing the Right Standard

There is no one "right" standard for implementing a MOSA in system development. Standards must be evaluated and selected based on their alignment with the MOSA objectives for a particular system. Program Managers (PMs) and systems engineers should understand the scope, purpose, and applicability of a standard or specification and how standardization supports their program's technical and business goals.

Ensuring that the right standards are available when they are needed often begins long before standards are implemented. Technology standardization begins with research and development, in what some have termed the "pre-standardization phase." This is particularly relevant for new and emerging technologies. When a program identifies a standardization need and no standard currently exists or the technology is rapidly evolving, the Department often supports acceleration of development of standards for a known technical gap. For example, an office may invest in research and development or participate in internal or external road-mapping efforts. American National Standards Institute–facilitated standardization collaboratives provide a forum for government and industry to identify gaps in standards in a particular domain and develop roadmaps for filling those gaps. Standards development takes time and resources, so identifying standardization needs early is essential to having the necessary standards available when they are needed.

When multiple standards exist, PMs and systems engineers must diligently conduct analyses or trade studies that compare among the various standards and choose those that best match their program's MOSA needs. Some considerations that may impact standardization decisions include the applicability of the standard (or its technological match) to the desired technical outcome, the program's schedule, the cost of standards conformance verification, and whether updates to new versions of a standard remain backward compatible. Ultimately, standardization decisions are deliberate choices that require trade-offs among program cost, schedule, and performance. An organization's business and technical objectives also drive standardization decisions for technology development.

Selecting standards that have achieved widespread adoption can maximize ability to implement a MOSA, though it is not guaranteed. When a standard or specification is more open, it is more likely that its use will help achieve more open modular system interfaces to support a MOSA. In contrast, using additional standards and specifications does not increase the chance of achieving a MOSA; the greater number of competing or overlapping standards can even hinder the ability to achieve a MOSA.

Company standards and specifications, along with company-limited stakeholder participation may result in instances of vendor lock, as modules are developed using proprietary information from standards that are not accessible to others beyond the company, preventing bids for modules from independent third-party vendors from being accepted. Using a MOSA and acquiring the appropriate technical baseline artifacts accompanied by appropriate IP rights to those artifacts should reduce the potential for vendor lock with respect to competing sustainment and upgrades for the program's remaining lifecycle.

Standardization in the Cell Phone Market

Until 2015, Android devices primarily used micro-Universal Serial Bus (USB) cables for charging, while Apple introduced its proprietary Lightning cables for iOS devices starting with the iPhone 5 in 2012. Lightning and micro-USB cables were not interchangeable, meaning an Android user could not use a Lightning cable to charge the device, and vice versa. Neither cable choice was inherently wrong, but the lack of standardization frustrated users and increased costs. In 2015, Android devices started transitioning to USB-C charging cables, and Apple followed suit in 2023. Standardizing to USB-C cellphone charging cables allowed for faster data transfer speeds, interchangeability, and higher power delivery. This evolution shows how standardization decisions can affect users and change over time.

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While government-unique standards are not prone to vendor lock, limits on participation and access to the standards may limit the benefits of standardization. Programs must balance critical information security and operational security requirements against the openness of the standards they develop and use when working with military capabilities. For example, defense standards and specifications may need to address controlled unclassified information or classified information. Decisions around government-unique standards must be based on program goals and should maximize openness while maintaining security.

Likewise, when choosing the appropriate standards or specifications to define interfaces, programs can consider international standardization agreements (ISAs) to achieve interoperability with allies and partners. Interoperability is one of the five core benefits of a MOSA. For example, programs can assess international standardization agreements (e.g., Standardization Agreements, Allied Engineering Publications), which are one of the four mandatory considerations for standardization.

When standards exist but do not meet the Department's needs, programs should consider adapting or revising the existing standard rather than developing an entirely new standard. DoDM 4120.24, *Defense Standardization Program (DSP) Procedures*, outlines several ways in which this can be achieved depending on the extent of the revision and relevance to commercial applications.

When a non-government standard is available, and the Department's requirements are relevant to commercial applications, the preferred option is to work with the standard development organization to revise and update the standard to meet the Department's needs. This may also be a feasible solution when working to meet a military-unique requirement. Alternatively, the Department may issue a defense specification which invokes the non-government standard and adds or amends only military-unique requirements.

If no standard exists, development of a non-government standard is preferred to military-unique requirements. *DoD Guidance on Participating in the Development and Use of Non-Government Standards* (SD-9) provides basic information about types of standards development organizations and best practices for engaging in non-government standards bodies.

4.1 When to Standardize

Defense Standardization Program (DSP) Procedures (DoDM 4120.24) provides guidance to program offices for decisions related to where, when, how, and to what level to standardize (outlined in Figure 6).

DoDM 4120.24 Standardization Considerations

- ✓ Is physical uniformity a minimum essential requirement?
- ✓ Is uniform configuration necessary for ease of operation or safety?
- ✓ Must form, fit, function, or interface be defined to permit interoperability or connectivity between discrete items?
- ✓ Will the item be used in a variety of applications?
- ✓ Is design control necessary because predictable performance is a minimum essential requirement (reliability, maintainability, survivability, and safety)?
- ✓ Must form, fit, function, or interface be identical to replace or substitute for an equivalent item (interchangeability)?
- ✓ Are there unique environmental requirements, such as radiation hardness and shock?
- ✓ Will there be economies of scale (consolidated buys)?
- ✓ Are there stocking and storage considerations (item will be in inventory)?
- ✓ Would training for operation, maintenance, or repair be improved by standardization?
- ✓ Would research and development costs, engineering time, or procurement time be reduced?

Figure 6. DoDM 4120.24 Standardization Considerations

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Many of these considerations and decisions are critical to standardization for MOSA. For example, “Must form, fit, function, or interface be defined to permit interoperability or connectivity between discrete items?” describes the inherent characteristics of MOSA systems design. Either of the two approaches outlined in 10 U.S.C. §4401 (and explained in Section 3.1 and Section 3.2 in this publication) can be used to help modular system interfaces achieve openness for MOSA. Further, programs should implement U.S.-ratified ISAs when 1) developing a module for foreign military sales or 2) considering integrating foreign-developed modules.

DoDM 4120.24 also provides guidance on when it may not be a good idea to standardize. Programs should avoid standardizing before a technology is stable and should not standardize in a way that limits design flexibility or innovation. Standardization in the context of a MOSA provides a design approach to addressing these limitations. While standardizing an unstable technology can be risky, within the context of a MOSA, standards may be used to encapsulate functional elements, turning each element into a “black box” that provides freedom for rapid technology insertion of advanced and new technologies over the lifecycle of a program. Similarly, within a MOSA, interface and standardization decisions should be made to optimize flexibility and innovation by defining severable module interfaces that allow for competition and the composition of new capabilities as technology evolves.

Finally, choosing the right standards should continue throughout the lifecycle of a program or system. As technology evolves and requirements change, standards and standardization needs must be revalidated. As the demand for a capability scales and as technology transitions into broader commercial use, programs may find it beneficial to transition company standards, defense standards, or consortia standards to a standards development organization with greater participation. This can present opportunities for increased production, more widespread support, and competition, thereby supporting the scalability, improved interoperability, and affordability of a system.

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Section 5. Criteria for MOSA-Enabling Standards

The decision tree in Figure 7 helps users identify and evaluate standards and specifications that can be used to enable a MOSA. The flowchart focuses on technical and business considerations for evaluating standards because a MOSA is a technical and business approach. Use this flowchart to evaluate big-picture considerations, guided by the following questions:

- What technical applications can the standard or specification be used for?
- Is the standard or specification relevant to DoW applications?
- Does the standard or specification enable any of the five major MOSA benefits?

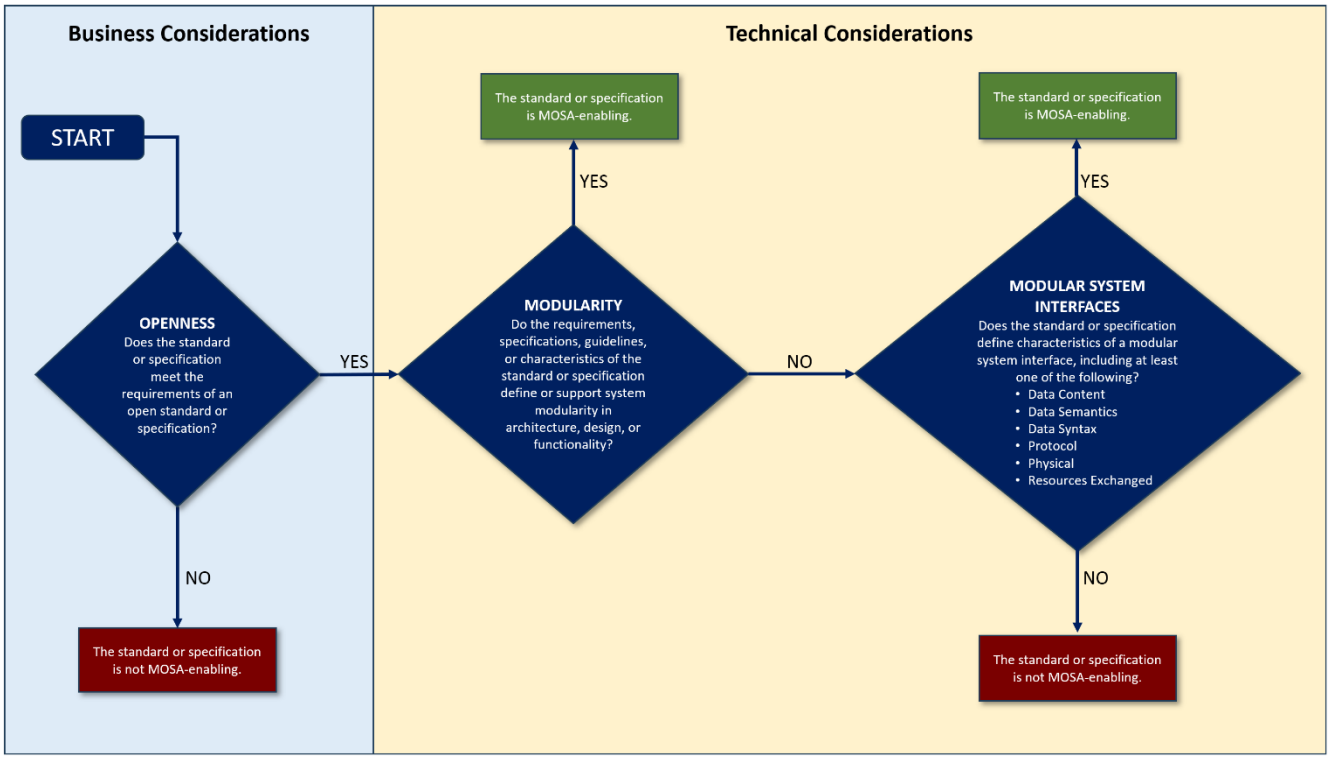


Figure 7. Criteria for the Identification and Evaluation of MOSA-Enabling Standards

5.1 Business Considerations

Business considerations within the flowchart are focused on the openness of a standard or specification. Realizing a MOSA requires that both industry and government stakeholders have access to standards and specifications, as well as the appropriate IP and licensing rights for use. Openness should be reviewed from the context of Figure 2.

5.2 Technical Considerations

To determine whether a standard or specification that meets the initial openness requirement has technical considerations for a MOSA consider its modularity and its use of MSIs.

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5.2.1 *Modularity*

The modularity diamond in Figure 7 is focused on the modularity of software and hardware-based components. Modularity is focused on the partitioning of software and hardware components into discrete, scalable, and self-contained functional elements so that they can be independently added, removed, or replaced without impacting the function of other components. Modularity and the use of modular systems can vary between DoW weapon systems. It is up to government stakeholders to identify the desired modularity in a system, and at what levels of indenture. Theoretically, the scale of modularity can occur from nuts, bolts, and fasteners all the way to system and system of systems components.

Examples of standards that would meet the modularity consideration include high-level MOSA architectural standards such as the Future Airborne Capability Environment, Sensor Open Systems Architecture, and Open Mission Systems.

5.2.2 *Modular System Interfaces*

The Modular System Interfaces diamond in Figure 7 is focused on the evaluation of a standard or specification based on its specific relevance to modular system interfaces. Government stakeholders generally determine where to locate the modular system interfaces, like desired modularity, within a system are based on programmatic and enterprise MOSA objectives. A holistic perspective to consider when evaluating standards based on this consideration is to optimally match the needs of the system or DoW application to the standards or specifications applicable to the system type or domain, mission needs, and logistics considerations.

The examples in Section 5.2.2.1 through Section 5.2.2.3 showcase how to use these criteria.

5.2.2.1 MIL-DTL-83133, Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34) and NATO F-35

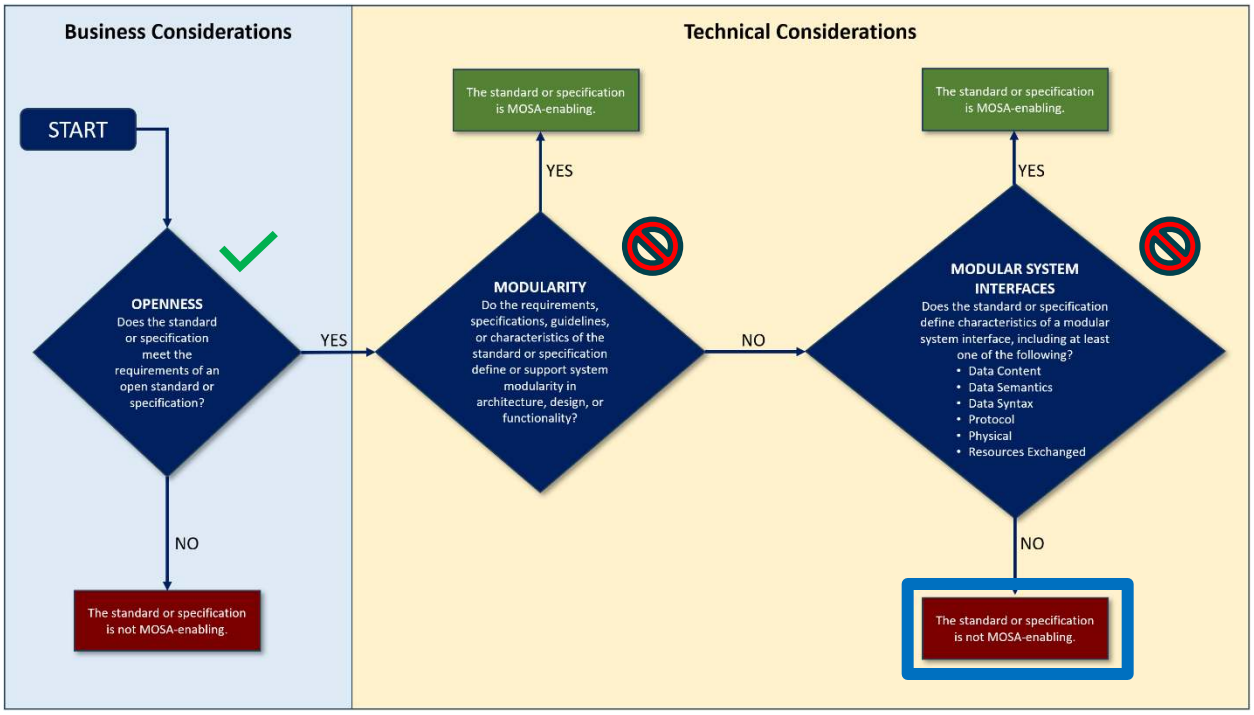


Figure 8. MIL-DTL-83133 Is Not MOSA-Enabling

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MIL-DTL-83133 meets the openness criteria because the detail specification is available on ASSIST, the official source of defense and federal standards and specifications, and is Distribution A. MIL-DTL-83133 does not invoke system modularity in architecture, design, and functionality on its own, nor does it define characteristics of a modular system interface. This specification is focused on the chemical composition of fuel, which invokes commonality of fuel used across aircraft systems. Therefore, as Figure 8 shows, MIL-DTL-83133 is not MOSA-enabling. Even though the detail specification has been ruled as not MOSA-enabling, it can be used in the development of modular systems.

5.2.2.2 Software Communications Architecture

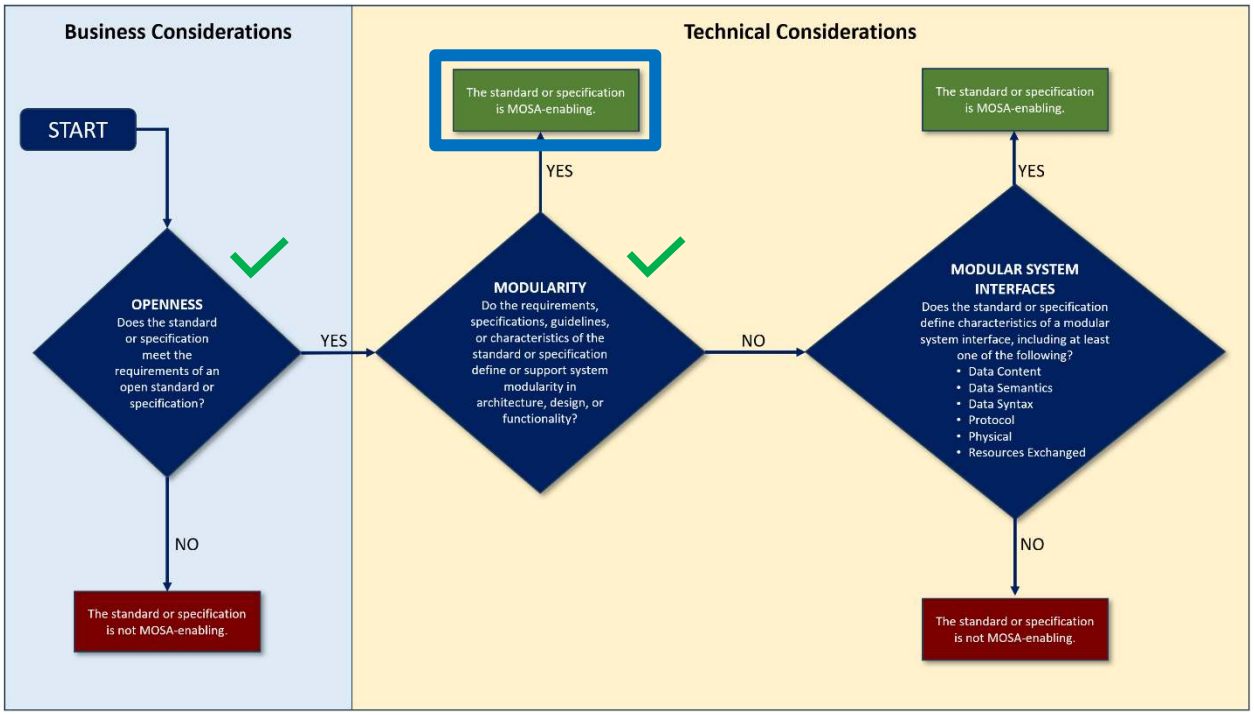


Figure 9. Software Communications Architecture Is MOSA-Enabling

The Software Communications Architecture (SCA) provides requirements for the development of software defined radio communication systems. SCA meets the openness criteria because the technical standard is published by the Joint Tactical Networking Center through partnership with industry partners via the Wireless Innovation Forum (i.e., government-industry collaboration). There is an adoption notice for SCA in ASSIST, indicating that the standard was adopted by DoW. As Figure 9 shows, SCA meets the requirements for providing modular design for enhanced interoperability of communication systems and reduces development and deployment costs. Therefore, SCA is MOSA-enabling.

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5.2.2.3 MIL-STD-3071, Tactical Microgrid Communications and Control

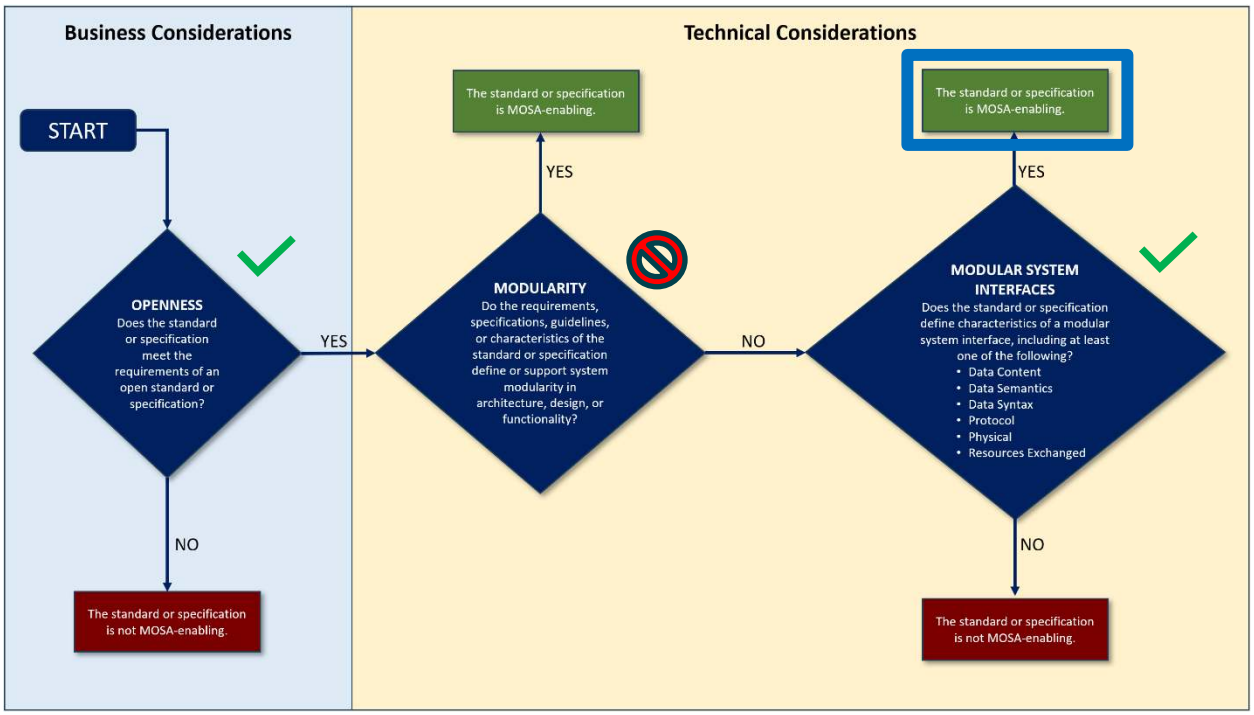


Figure 10. MIL-STD-3071 Is MOSA-Enabling

MIL-STD-3071 meets the openness criteria because the standard is available on ASSIST and is Distribution A. MIL-STD-3071 is an interface standard and establishes communication and control interface requirements for tactical microgrids (TMGs) to enable TMG components to operate as a single entity providing electricity in a tactical environment. As Figure 10 shows, the standard meets the requirements of the modular system interface diamond by defining component hardware and software interfaces, providing communication standards including messages for exchange between devices in the microgrid, and allowing plug-and-play capabilities for various equipment.

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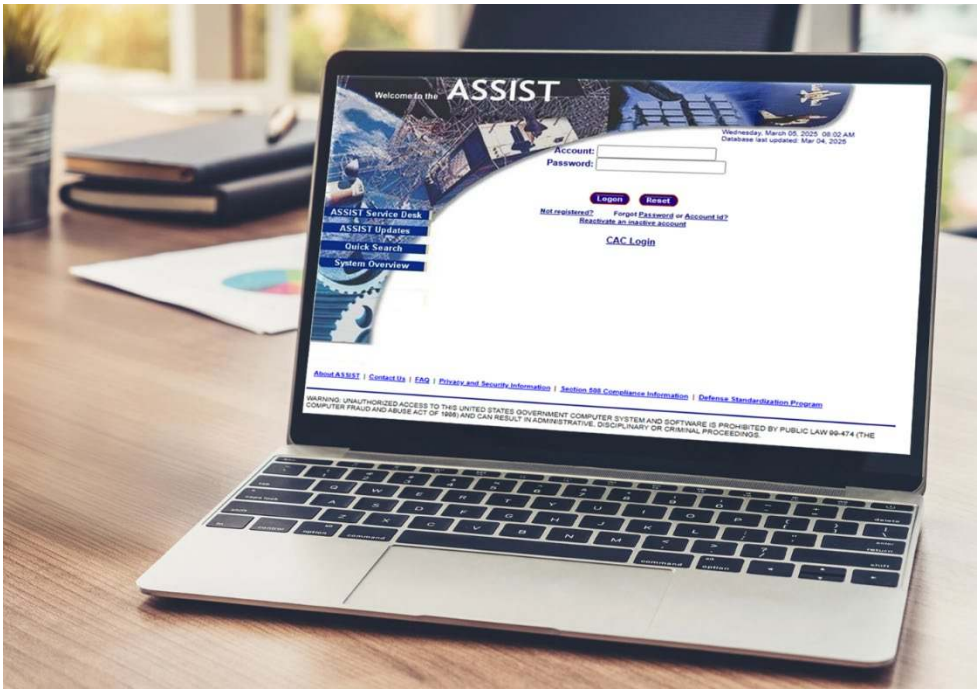
Section 6. MOSS Portfolio

Located in ASSIST (https://assistca.dla.mil/online/doc_mgt/mosa.cfm), the Modular Open Systems Standards and Specifications (MOSS) portfolio serves as the authoritative source for MOSA-enabling standards and specifications used by the DoW. This portfolio provides visibility into standards and specifications that can enable MOSA and hosts such standards in a singular location that enables enterprise-level access to the defense industry. It is not meant to prescribe standards the DoW must use for weapon systems.

PMs and engineers alike can use the portfolio during the architecting phase to determine how best to employ a modular design in weapon system architecture and what types of standards and specifications may be of interest in particular technical domains. Using the standards and specifications in this portfolio can help promote standards reuse within the DoW and prevent future duplicative MOSA standardization efforts.

The MOSS portfolio contains standards and specifications that enable MOSA. There is a two-pronged approach to capture open system standards and other standards that were not developed specifically to address MOSA but could be useful in the implementation of a MOSA. Open system standards can be found in the MOSS standardization area. Standards that can enable MOSA are tagged as MOSA-enabling in ASSIST, which encapsulates current standards (with justified return on investment) and future potential standards. The MOSS portfolio is a culmination of both approaches and is captured as a report in ASSIST.

This portfolio is continually updated, with the MOSWG Standards Tiger Team serving as the control board for changes. Though the use of MOSA-enabling standards alone does not guarantee MOSA implementation in warfighting systems, standards and specifications are one of the building blocks for applying a MOSA. They provide a means to validate that technology is safe, can be universally adopted, and is interoperable. Continued efforts to develop, identify, and mature MOSA-enabling standards will drive opportunities for the DoW to realize the five benefits of MOSA.



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Appendix A. Definitions

Term	Definition	Source
Company Standard	A company document that establishes engineering and technical limitations and applications for items, materials, processes, methods, designs, and engineering practices unique to that company.	MIL-HDBK-539v
Consensus-Based	General agreement, but not necessarily unanimity. During the development of consensus, comments and objections are considered using fair, impartial, open, and transparent processes.	OMB Circular A-119
Cybersecurity	Prevention of damage to, protection of, and restoration of computers, electronic communications systems, electronic communications services, wire communication, and electronic communication, including information contained therein, to ensure its availability, integrity, authentication, confidentiality, and nonrepudiation.	DoDI 8500.01
De Facto Standard	Something that has gained widespread acceptance in the marketplace but is not “officially” sanctioned by any international, national, or government organization.	DoDM 4120.24
Government-Unique Standard	A standard developed by the government for its own purposes.	OMB Circular A-119
Interoperability	The ability of systems, units, or forces to provide data, information, materiel, and services to, and accept the same from, other systems, units, or forces, and to use the data, information, materiel, and services exchanged to enable them to operate effectively together. IT interoperability includes both the technical exchange of information and the end-to-end operational effectiveness of that exchange of information as required for mission accomplishment in its operational environment including appropriate cybersecurity aspects. Interoperability is more than just information exchange. It includes systems, processes, procedures, organizations, and missions in appropriately stressed operational environments over the system’s lifecycle.	DoDM 4120.24
Machine-Readable Format	A format that can be easily processed by a computer without human intervention.	FY21 NDAA Section 804
Modular System Interface	A shared boundary between major systems, major system components, or modular systems, defined by various physical, logical, and functional characteristics, such as electrical, mechanical, fluidic, optical, radio frequency, data, networking, or software elements.	10 U.S.C. §4401-4403
Non-Government Standard	A national or international standardization document developed by a private sector association, organization, or technical society that plans, develops, establishes, or coordinates standards, specifications, handbooks, or related documents. This term does not include standards of individual companies. NGSs adopted by the DoD are listed in the ASSIST.	DoDM 4120.24
Semantics	The intended meaning of acceptable sentences of a language.	NIST SP 800-152

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Specification	A document prepared to support acquisition that describes the essential technical requirements for purchased materiel and the criteria for determining whether those requirements are met.	DoDM 4120.24
Standard	A document that establishes uniform engineering or technical criteria, methods, processes, and practices.	DoDM 4120.24
Syntax	The rules for constructing or recognizing the acceptable sentences of a language.	NIST SP 800-152

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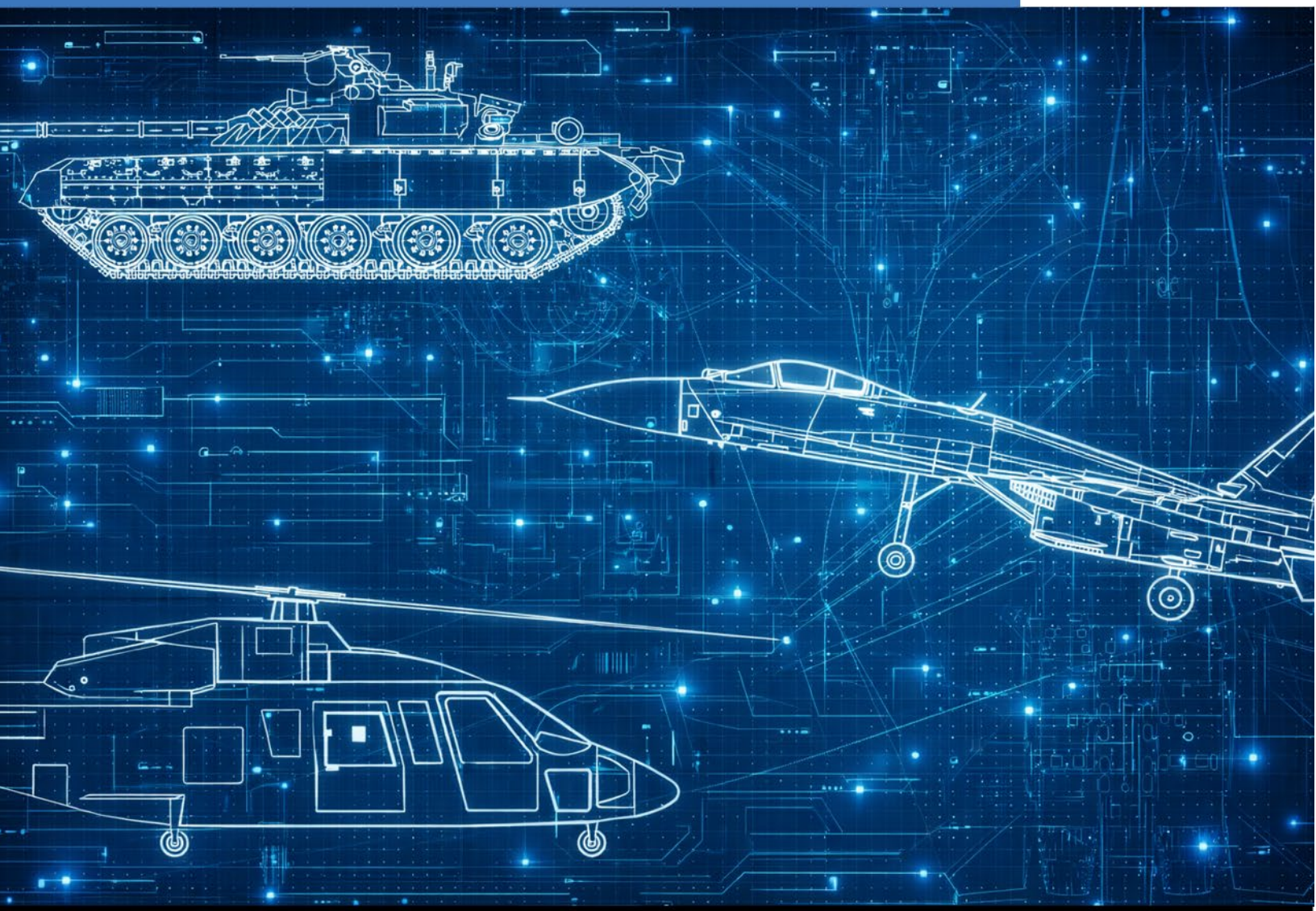
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Appendix C. Abbreviations

API	Application Programming Interface
DoDI	Department of Defense Instruction
DoDM	Department of Defense Manual
FY	Fiscal Year
IP	Intellectual Property
ISO/IEC	International Organization for Standardization/International Electrotechnical Commission
MIL-DTL	Military Detail
MIL-SPEC	Military Specification
MIL-STD	Military Standard
MOSA	Modular Open Systems Approach
MOSS	Modular Open Systems Standards and Specifications
MOSWG	Modular Open Systems Working Group
MSI	Modular System Interface
NATO	North Atlantic Treaty Organization
NGS	Non-Government Standard
OMB	Office of Management and Budget
PDF	Portable Document Format
PM	Program Manager
SD	Standardization-related Document
SDO	Standards Development Organization
SMA	Standards Management Activity
SysML	Systems Modeling Language
UML	Unified Modeling Language
U.S.C.	United States Code
USG	U.S. Government
XML	Extensible Markup Language

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